

SESSION-02
SECURITIES ANALYSIS & PORTFOLIO MANAGEMENT

From Return Measurement to Portfolio Pricing

A Capital Market Theory Journey



SESSION 1

What did I earn?

Return & risk measurement



SESSION 2

How do assets interact?

Covariance, correlation, beta, alpha



SESSION 3

What's the best combination?

Portfolio risk-return & diversification



SESSION 4

Where's the efficient boundary?

Efficient frontier, MVP, CML



SESSION 5

What should I be paid for risk?

CAPM & SML



SESSION 6

Did my manager add value?

Risk-adjusted performance evaluation

"By the end of these 6 sessions, you should be able to open any mutual fund fact sheet and know exactly which of these six questions its marketing is quietly avoiding. Which one would you guess — and why?"



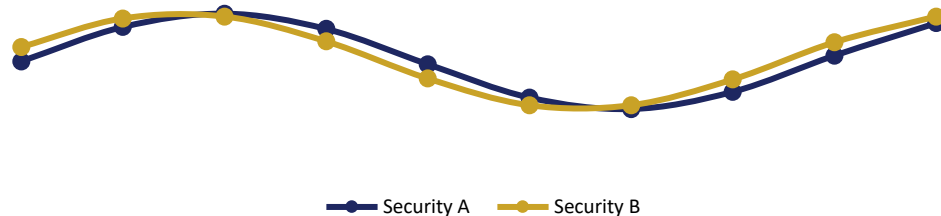
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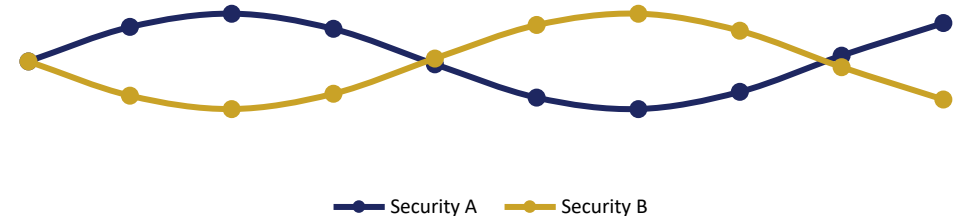
From Single Assets to Pairs: Why Interaction, Not Just Individual Risk, Matters

- Session 1 measured return and risk for one asset at a time — the new question: how do two return series move relative to each other?
- A “safe” (low-SD) asset can combine with another “safe” asset to create a risky portfolio — and the reverse is also true
- This session builds the vocabulary needed before any portfolio can be constructed

MOVING TOGETHER (POSITIVELY CORRELATED)



MOVING APART (NEGATIVELY CORRELATED)



Same individual volatility — very different combined behaviour.

CUE

“Can you name two Indian stocks you'd expect to move almost identically, and two you'd expect to move in opposite directions? What's driving each pair?”

QUESTION BANK LINK

None — framing slide. Sets up the covariance formula, which follows immediately.

Covariance: Measuring Co-Movement Between a Security and the Market

- $Cov(R_i, R_m) = \Sigma[(R_i - R_i)(R_m - R_m)] / n$
- Positive covariance → security and market move together | Negative → move apart
- Covariance is the single term separating portfolio risk from a simple weighted average of individual risks
- Every modern optimizer (mean-variance, Black-Litterman) uses a covariance matrix as its core input

Year	Dev. Stock A	Dev. Stock B	Product
Y1	+8	+6	48
Y2	-4	-3	12
Y3	+4	+5	20

Illustrative 3-year sample — products sum to 80.01, ÷ n = 3 → Cov = 26.67

COV(A,B)

26.67



Positive — A and B move together

CUE

“Reliance and Nifty have a covariance that's strongly positive. Gold and equity, during a crisis, can turn negative. What does each tell you about diversification benefit — and does the sign matter more than the magnitude?”

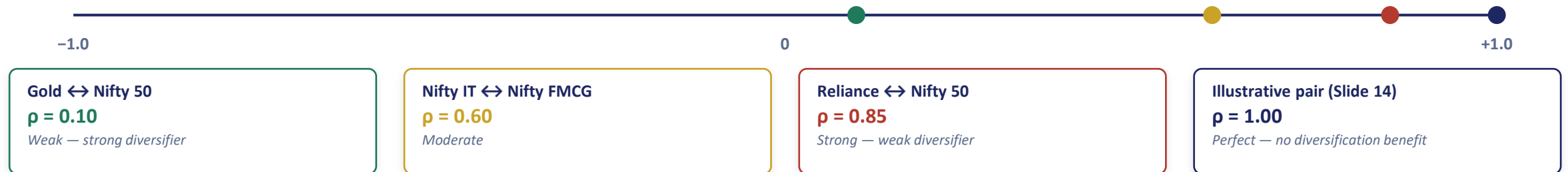
QUESTION BANK LINK

Use in class: Problem 4.3 (covariance, Security A & market)

Assign for practice: Problems 4.1, 4.2 (prerequisite average return & variance/SD)

Correlation Coefficient: Standardizing Covariance for Comparability

- $\rho(A,B) = \text{Cov}(A,B) / (\sigma_a \times \sigma_B)$ — range -1 to +1, scale-free across any asset class
- From Slide 14's inputs: $\rho = 26.67 / (5.7 \times 4.6) = +1.0$ (illustrative)
- Correlation, not covariance, is what portfolio managers actually quote and compare



CUE

“Why do portfolio managers quote correlation rather than covariance when comparing an equity fund to a gold ETF or a bond fund? What would go wrong if they compared raw covariances instead?”

QUESTION BANK LINK

Use in class: Problem 4.4 (correlation, Security A & market)

Total Risk Decomposition: Systematic vs Unsystematic Risk

- Total risk = Systematic risk (market-wide, non-diversifiable) + Unsystematic risk (firm-specific, diversifiable)
- Systematic: interest rates, inflation, GDP, geopolitical shocks — no portfolio escapes it
- Unsystematic: management change, product recall, factory fire, governance issues — eliminable through diversification
- “Finance's free lunch”: eliminating unsystematic risk costs nothing in expected return — it earns no risk premium

TOTAL RISK



Illustrative split — worked from Question Bank Problem 4.8

Market does not compensate for risk you could have **diversified away for free**.

Unsystematic risk earns **no risk premium** — it is eliminable at zero cost.

CUE

“Concentrated, promoter-driven holdings are extremely common among Indian retail investors. Is that unsystematic risk being taken knowingly — or is it a diversification failure dressed up as conviction?”

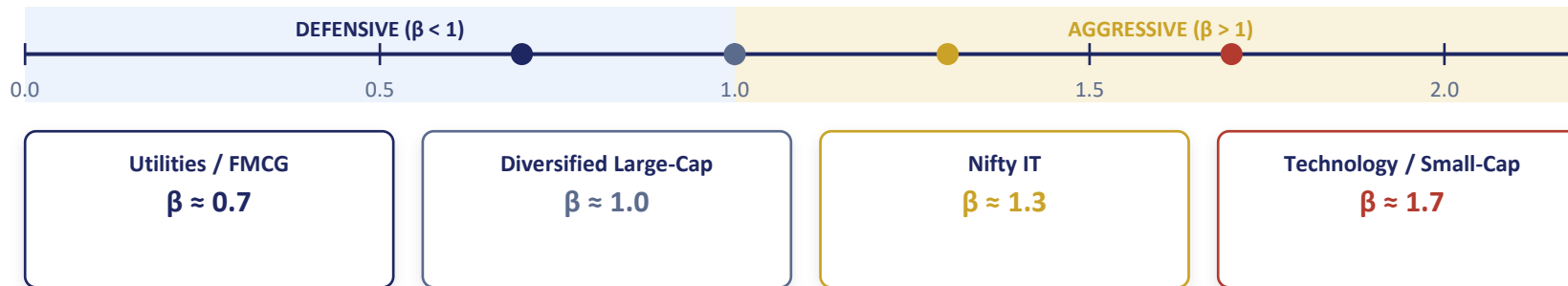
QUESTION BANK LINK

Use in class: Problem 4.8 (percentage risk decomposition)

Assign for practice: Problem 4.9 (risk decomposition in SD terms)

Beta: Measuring Market Sensitivity

- $\beta = \text{Cov}(R_i, R_m) / \text{Var}(R_m) = \text{Corr}(R_i, R_m) \times (\sigma_i / \sigma_m)$
- $\beta > 1$ Aggressive | $\beta = 1$ Market-like | $\beta < 1$ Defensive | $\beta = 0$ No relationship | $\beta < 0$ Counter-cyclical
- Caution: individual-stock betas drift toward 1.0 over time (Blume, 1971); portfolio-level betas are far more stable



WORKED EXAMPLE

$$\beta = 0.80 \times (18/12) = 1.20$$

Nifty IT ($\beta \approx 1.3$) vs Nifty FMCG ($\beta \approx 0.7$) — same economy, very different sensitivity.

“Is a stock with $\beta = 1.5$ always riskier than one with $\beta = 0.6$? What if the low-beta stock carries enormous unsystematic business risk instead?”

QUESTION BANK LINK

Use in class: Problem 4.5 (beta of Security A)

The Great Beta Debate: Does Beta Really Measure Risk?

- An investment-committee debate, not a settled question — no resolution offered yet
- This argument returns with harder empirical evidence in Session 5 (Roll's Critique, anomalies)



ADVOCATES

- Theoretically elegant — single market-risk factor
- Widely used in cost-of-equity and M&A valuation
- Regulator-accepted (SEBI, SEC)
- Forward-looking if estimated correctly



CRITICS

- Backward-looking — historical data for a future decision
- Fama-French: size and value explain returns better than beta alone
- Low-volatility anomaly: low-beta stocks persistently outperform
- Buffett: Berkshire's low beta, extraordinary long-run returns

CUE

“Can you name a successful long-term investor whose track record is primarily explained by taking on high beta? Why is that example so hard to find?”

QUESTION BANK LINK

None — conceptual debate slide.

CAPM Expected Return and Security Alpha

- CAPM: $E(R_i) = R_f + \beta_i(R_m - R_f)$ — the return a security should earn given its systematic risk
- Alpha: $\alpha = \text{Actual Return} - \text{CAPM Required Return}$
- SPIVA India: a large majority of active large-cap funds underperform their benchmark over 5 years — “where did the alpha go?”



Actual return of 19% against a CAPM-required return of 15.6% $\rightarrow \alpha = 19\% - 15.6\% = +3.4\%$ (outperformance after adjusting for risk)

CUE

“If markets are semi-strong efficient, alpha shouldn't persist. So where does measured alpha actually come from — and how long can any single source of it survive before it gets arbitrated away?”

QUESTION BANK LINK

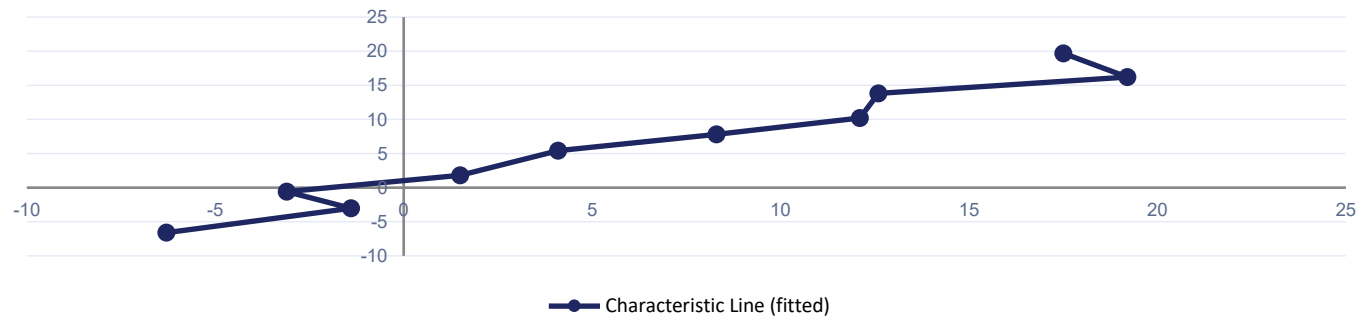
Use in class: Problem 4.6 (CAPM required return, Security A — the alpha benchmark)

⚠ Flag: Problem 4.7 (“Alpha of Security A”) is missing from the source Question Bank between 4.6 and 4.8. Use the illustrative $\alpha = +3.4\%$ example above until restored.

The Characteristic Line: The Regression Behind Beta and Alpha

(Special, complete slide — the integration point for this session's covariance, correlation, beta, and alpha problems)

- $R_i = \alpha_i + \beta_i R_m + \varepsilon_i$ — the regression of a security's return on the market's return
- Security return (R_i) is the dependent variable; market return (R_m) is the explanatory variable
- The intercept is alpha; the slope is beta. β captures the systematic component; the residual (ε) is unsystematic risk
- Beta and alpha are never directly observed — always statistically estimated from a chosen index, frequency, and sample window



Intercept = α

Where the line crosses R_i when $R_m = 0$

Slope = β

How steeply security return responds to market return

CUE

“Two analysts estimate beta for the same stock — one uses Nifty 50 as the benchmark, the other a global index; one uses monthly returns, the other daily; one uses 3 years of data, the other 5. Should they get the same beta? What does it mean that they usually don't?”

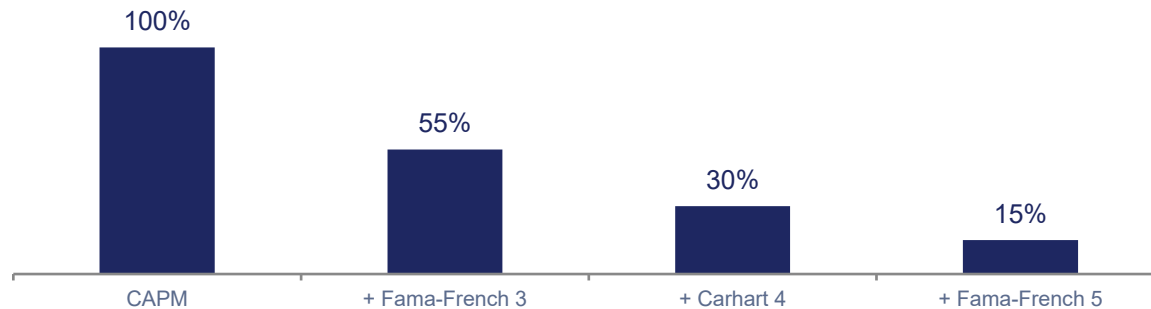
QUESTION BANK LINK

Integration point for Problems 4.3, 4.4, 4.5, 4.6 (and 4.7, subject to the Slide 19 content-gap flag) — every prior problem this session is an application of this one regression.

Skill, Luck, or Factor Exposure? Why Most Alpha Disappears

- CAPM (1964) → Fama-French 3-Factor (1993, +Size +Value) → Carhart 4-Factor (1997, +Momentum) → Fama-French 5-Factor (2015, +Profitability +Investment)
- Each step reclassifies more measured “alpha” as systematic factor exposure, not genuine skill
- Berk & Green (2004): only ≈2–3% true alpha survives in even the best funds after full factor-adjustment

SHARE OF “ALPHA” STILL UNEXPLAINED (ILLUSTRATIVE)



✓ SKILL INDICATORS

Persistent 5+ years · survives factor adjustment · holds in bull & bear markets · capacity-constrained (low AUM)

✗ LUCK INDICATORS

Recent-only · disappears after factor adjustment · one concentrated bet · rapidly growing AUM

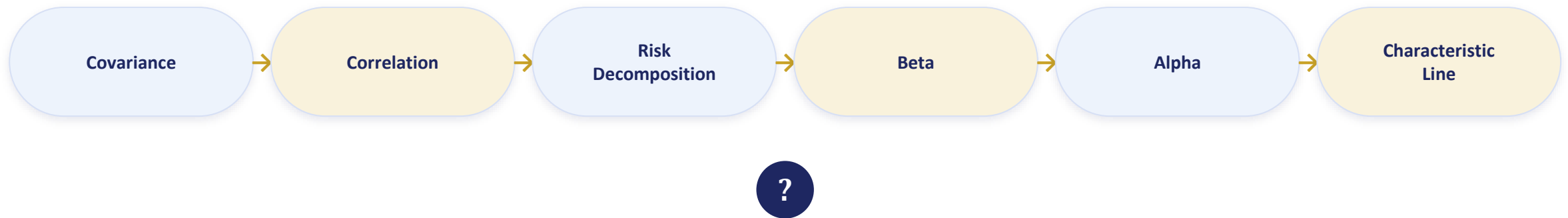
“Are star fund managers genuinely skilled, or are they simply ‘lever-pressing’ on systematic factors that happen to be in favour? Does the answer change how you’d allocate capital to them?”

QUESTION BANK LINK

None — conceptual extension of Slide 19’s alpha discussion.

Session 2 Synthesis: From Security-Level Metrics to Portfolio Construction

- Every metric so far describes ONE security against the market — not yet a multi-asset portfolio
- CAPM and alpha assume you already know how to combine assets efficiently — Session 3 supplies that missing piece
- Beta debate and alpha-vs-factor-exposure questions remain open — both return with harder evidence in Session 5



Every metric so far describes ONE security against the market.

Session 3 → Portfolio Risk-Return Framework, Diversification & Markowitz Theory

“Every metric in this session describes ONE security against the market. What's still missing to build an actual multi-asset portfolio?”

QUESTION BANK LINK

None — pure recap/bridge slide closing Session 2.